

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T43 · Area and Perimeter

Questions

Student book · questions with writing space

■ 16 pages

FAMILY MAP

Area Perimeter

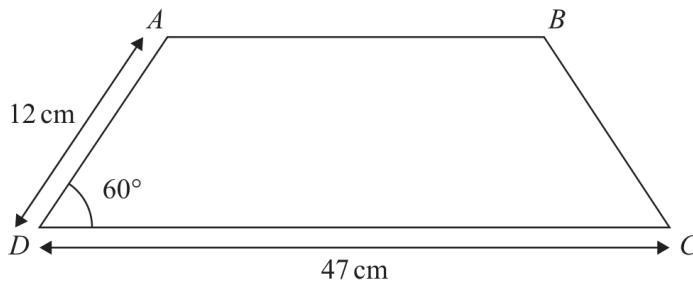
F01	Use parallelogram and trapezium area	5 Q 19 marks
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F02	Use rectangle and triangle area	7 Q 32 marks
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F03	Find perimeter of polygonal shapes	3 Q 11 marks
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Working Unit 06

4MA1-1H Q9

UNIT 1
5 marks**9**Diagram **NOT**
accurately drawn $ABCD$ is a trapezium with one line of symmetry.angle $ADC = 60^\circ$ $AD = 12 \text{ cm}$ $DC = 47 \text{ cm}$

Work out the area of the trapezium.

Give your answer correct to 3 significant figures.

Show your working clearly.

..... cm^2 **(Total for Question 9 is 5 marks)**

Question 21

4MA1-2H Q7

MODULAR UNIT 1

4 marks

- 7 The diagram shows a 6-sided shape $ABCDEF$

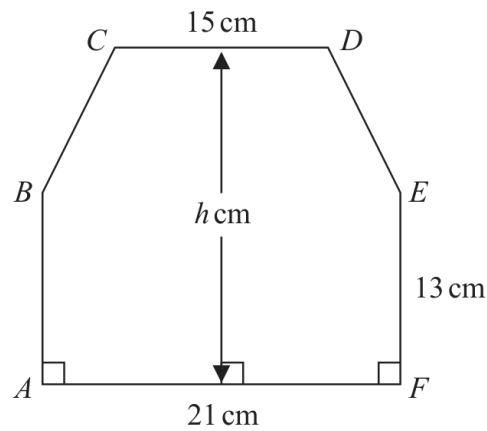


Diagram **NOT**
accurately drawn

$$AF = 21 \text{ cm} \quad CD = 15 \text{ cm} \quad AB = FE = 13 \text{ cm}$$

The perpendicular height of the shape is h cm

CD is parallel to AF

The area of the shape is 390 cm^2

Work out the value of h

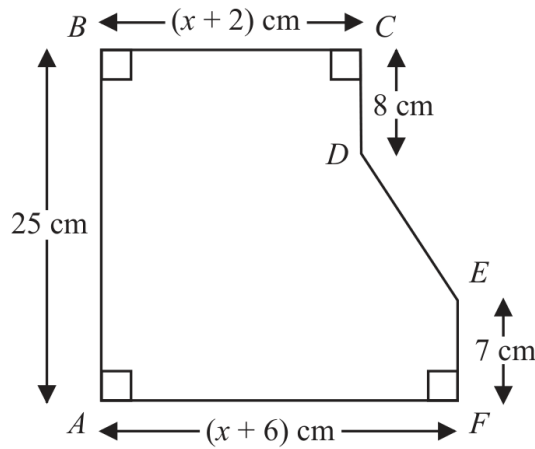
$h = \dots\dots\dots$

Question 22

4MA1-2H Q10 whole

MODULAR UNIT 1

5 marks

10 The diagram shows a hexagon $ABCDEF$ Diagram **NOT**
accurately drawn

$$AB = 25 \text{ cm} \quad BC = (x + 2) \text{ cm} \quad CD = 8 \text{ cm} \quad EF = 7 \text{ cm} \quad AF = (x + 6) \text{ cm}$$

The area of hexagon $ABCDEF$ is 258 cm^2 Work out the value of x $x = \dots\dots\dots$

Question 18

4MA1-2H Q5 Q5

MODULAR UNIT 1

4 marks

- 5 The diagram shows an 8-sided shape $ABCDEFGH$.

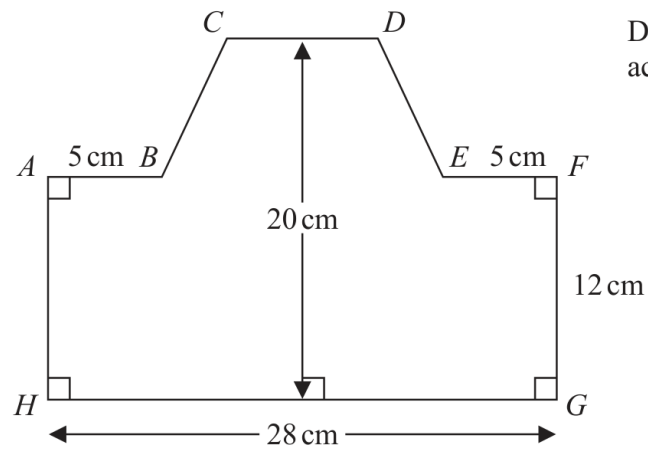


Diagram **NOT**
accurately drawn

$$HG = 28 \text{ cm} \quad FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

CD is parallel to HG

The area of shape $ABCDEFGH$ is 434 cm^2

Find the length of CD .

..... cm

Question 07

4MA1-1H Q9

MODULAR UNIT 1

4 marks

- 9 The diagram shows isosceles triangle ABC

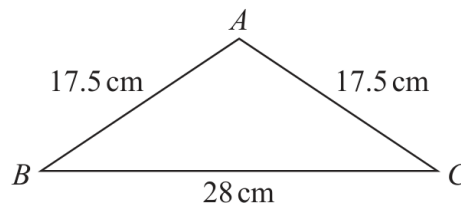


Diagram **NOT**
accurately drawn

$$AB = AC = 17.5 \text{ cm}$$

$$BC = 28 \text{ cm}$$

Calculate the area of triangle ABC

..... cm^2

Diagram **NOT**
accurately drawn

Work out the value of x .

14 The diagram shows parallelogram $EFGH$.

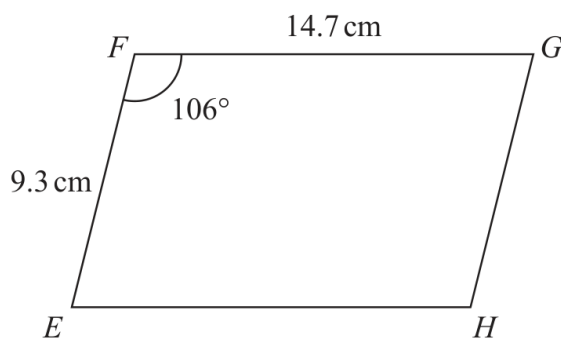


Diagram **NOT**
accurately drawn

$$EF = 9.3 \text{ cm}$$

$$FG = 14.7 \text{ cm}$$

$$\text{Angle } EFG = 106^\circ$$

- (a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm^2
(2)

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

20

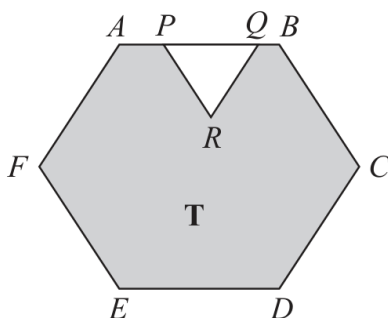


Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

work out the length of PQ .

..... cm

Work out the value of x .

-
- The diagram shows a composite figure. It consists of a rectangle with a height of 2 m and a width of 0.75 m. On top of the rectangle, there is a semicircle with a diameter of 0.5 m. The semicircle is centered on the top edge of the rectangle.

..... millilitres

10 ABC is an isosceles triangle with $BA = BC$.

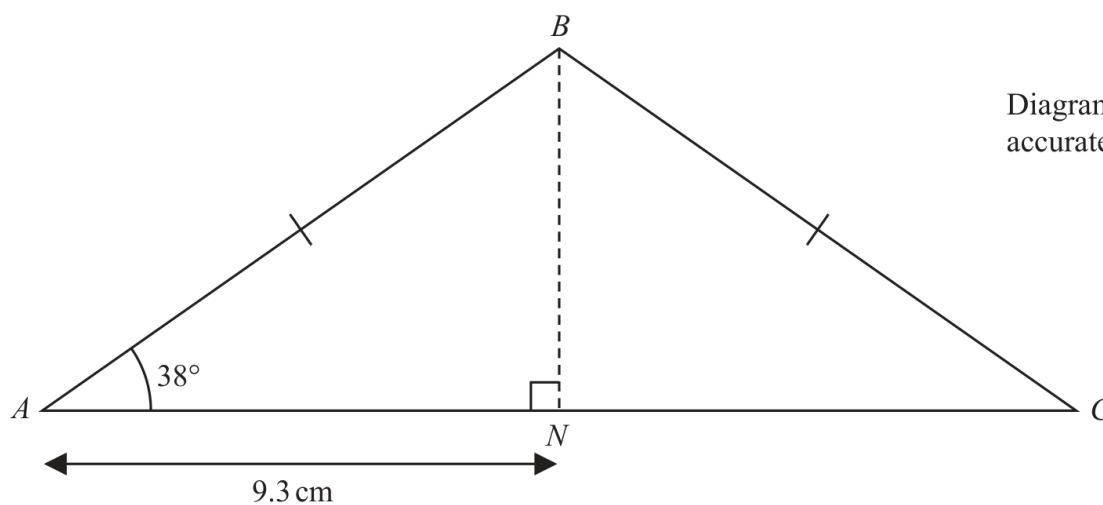


Diagram **NOT**
accurately drawn

N is the point on AC such that $AN = 9.3$ cm and BN is perpendicular to AC .

Work out the perimeter of triangle ABC .

Give your answer correct to 3 significant figures.

..... cm

-
- A trapezoid $ABCD$ is shown with parallel bases BC and AD . The length of base BC is 8.4 cm and the length of base AD is 17.6 cm .

Work out the size of angle ABC .
Give your answer correct to 1 decimal place.

- 8 The diagram shows an isosceles triangle.

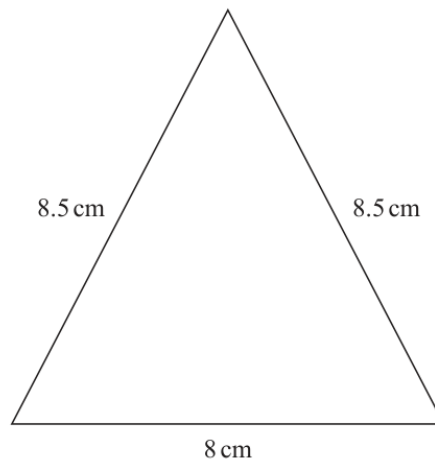


Diagram **NOT**
accurately drawn

Work out the area of the triangle.

.....cm²

- 4 The diagram shows rectangle $ABCD$

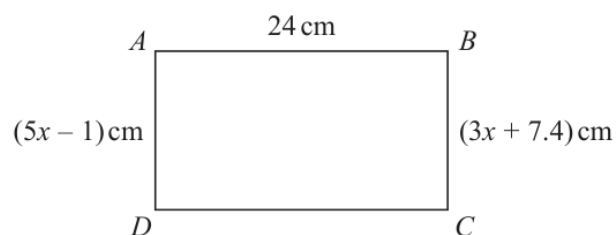


Diagram **NOT**
accurately drawn

Work out the perimeter of the rectangle.
Show your working clearly.

..... cm